Kibana, and we play with Elasticsearch by indexing a few documents and running search queries and analytics. Stay tuned!

Summary

- Search is the new normal and the most sought-after functionality for organizations, enabling competitive advantage.
- Search engines built using relational databases as the backend used to serve our search purposes but can't fulfill the full-fledged search functionality found in modern search engines.
- Modern search engines provide multifaceted, full-text search capabilities and multifold benefits from basic search to advanced search and analytical functions, all with split-second performance. They are also expected to handle terabytes to petabytes of data and scale if needed.
- Elasticsearch is an open source search and analytics engine built over Apache Lucene. It is a highly available server-side application developed in Java.
- Because Elasticsearch was designed as a programming-language-agnostic product, communication with the server takes place over HTTP using rich RESTful APIs. These APIs receive and send data in JSON format.
- The Elastic Stack is a suite of products composed of Beats, Logstash, Elasticsearch, and Kibana. Beats are single-purpose data shippers, Logstash is a data-processing ETL (extract, transform, load) engine, Kibana is the administrative UI tool, and Elasticsearch is the heart and soul of the stack.
- The Elastic Stack enables an organization to position itself in three core areas: search, observability, and security.
- Elasticsearch has become popular over the last few years due to its structured/unstructured search and analytics capabilities; rich set of RESTful APIs; its schema-free nature; and performance, high-availability, and scalability characteristics.
- AI-powered search is here. With the advent of generative AI and ChatGPT, the search space will be explored further, and search will become more intuitive and predictive.